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RESEARCH ARTICLE

Elevated methylmercury production in mercury-contaminated soil and its bioaccumulation in rice: key roles of algal decomposition

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HIGHLIGHTS

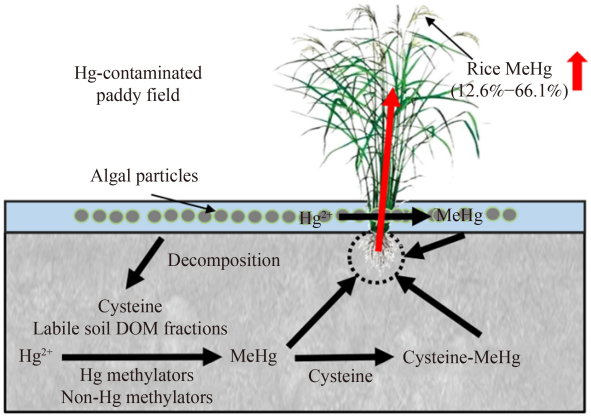
- AOM input elevates water-soluble cysteine and labile DOM fractions in soil.
- AOM input fuels potential Hg methylators and non-Hg methylators in soil.
- Decayed algal aggregate is Hg methylating "hotspot" and MeHg source in soil.
- AOM-driven SDOM variations elevate soil MeHg production and bioaccumulation in rice.

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GRAPHIC ABSTRACT



ABSTRACT

Algal-derived organic matter (AOM) regulates methylmercury (MeHg) fate in aquatic ecosystems, whereas its role in MeHg production and bioaccumulation in Hg-contaminated paddies is unclear. Pot and microcosm experiments were thus performed to understand the response characteristics of MeHg concentrations in soil and rice in different rice-growing periods to algal decomposition. Compared to the control, algal decomposition significantly increased soil water-soluble cysteine concentrations during the rice-tillering and grain-filling periods ($P < 0.05$). It also significantly lowered the molecular weight of soil-dissolved organic matter (SDOM) during the rice-tillering period ($P < 0.05$) and SDOM humification/aromaticity during the grain-filling period. Compared to the control, AOM input increased the abundance of potential Hg and non-Hg methylators in soil. Furthermore, it also greatly increased soil MeHg concentrations by 25.6%–80.2% and 12.6%–66.1% during the rice-tillering and grain-filling periods, with an average of 42.25% and 38.42%, respectively, which were significantly related to the elevated cysteine in soil and the decrease in SDOM molecular weight ($P < 0.01$). In the early stage (within 10 days of microcosm experiments), the MeHg concentrations in decayed algal particles showed a great decrease ($P < 0.01$), suggesting a potential MeHg source in soil. Ultimately, algal decomposition greatly increased the MeHg concentrations and bioaccumulation factors in rice grains, by 72.30% and 16.77%, respectively. Overall, algal decomposition in Hg-contaminated paddies is a non-negligible factor promoting MeHg accumulation in soil-rice systems.

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1 Introduction

Mercury (Hg) is of global concern as it can be converted

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into highly toxic methylmercury (MeHg) and suffers from considerable biomagnification in food chains (Ullrich et al., 2001; Liu et al., 2019). As Hg methylation is predominantly conducted by anaerobic microorganisms, paddy fields are an important potential Hg-methylating "hotspot" (Ma et al., 2019). Rice (*Oryza sativa* L.) has a high MeHg accumulation capacity, especially in Hg-mining areas, where rice MeHg concentration was found to be 10–100 times higher than in other crops (Zhang et al., 2010; Meng et al., 2011). Rice is a staple food worldwide, especially in Asia, but a growing body of evidence suggests that rice consumption is becoming an important dietary source of MeHg exposure for humans (Liu et al., 2019a), presenting a major health threat.

There are many factors controlling MeHg production in soil and its bioaccumulation in rice (Zhao et al., 2020). Among these, soil-dissolved organic matter (SDOM), an extremely complex component with a low to high molecular weight, is of particular importance (Ravichandran, 2004; Fang et al., 2011). It either serves as a metal-ligand promoting or limiting Hg bioavailability or as a bioavailable carbon source fueling Hg-methylating bacteria, thus greatly controlling MeHg production (He et al., 2019). Except for dissolved organic matter (DOM) concentration, its molecular properties and sources provided more information to understand Hg-DOM interactions (Bravo et al., 2017; He et al., 2019; Yang et al., 2022). For example, low-molecular-weight (LMW) compounds can facilitate the microbial Hg uptake to promote intracellular MeHg synthesis or act as important MeHg translocation assistors, favoring the MeHg uptake and translocation of MeHg in rice (Schaefer and Morel, 2009; Zhao et al., 2018; Hao et al., 2022). Additionally, organic matter (OM) derived from phytoplankton can result in higher microbial MeHg production than terrigenous OM in the lakes (Bravo et al., 2017). In the paddy fields, crop straws are the important plant-derived OM in soil (Xiong, 2023), which are found to greatly increase microbial MeHg production and bioaccumulation of MeHg in rice (Tang et al., 2019). The interaction between crop straw-released DOM and Hg in soil determines MeHg production and bioaccumulation, which are well documented in recent studies (Tang et al., 2019; Su et al., 2021). However, other plant-derived OMs, such as algae, are less focused, although it is also an important carbon regular in the paddy fields (Wang et al., 2022).

In the paddy fields, agricultural activities such as regular fertilization result in algal blooms, which are estimated to range from 50 to 620 kg/ha in biomass (Li et al., 2020; Wang et al., 2022). Algae inhabiting the soil-water interface in paddy fields accumulate organic carbon during the growing period, which is then released after their death, contributing to much of the algal-derived OM (AOM) (Wu et al., 2016; Li et al., 2020; Wang et al., 2022). Recent evidence has linked AOM to the elevated MeHg production in lake sediments and its bioaccumulation in aquatic organisms (Pickhardt et al., 2005; Bravo et al., 2017; Lei et al., 2021; Li et al., 2022; Wei et al., 2022). Algae can release abundant LMW AOM components (e.g., cysteine) via cell metabolism and decomposition, fueling the microbial Hg methylators inhabiting extracellular polymeric substances or lake sediments (Leclerc et al., 2015; Lei et al., 2019; 2021; Zhang et al., 2022). These LMW thiol-containing compounds also have a high affinity for Hg or MeHg, increasing Hg bioavailability (Lázaro et al., 2019; Mangal et al., 2019). In addition, the deposition of algal particles could contribute to microbial Hg methylators in water or lake sediments, promoting MeHg production (Lei et al., 2019; Xiang et al., 2021). Although these findings suggest that AOM input may significantly affect MeHg production in soil and its bioaccumulation in rice in Hg-contaminated paddy fields, limited information has been published on this subject.

In this study, we hypothesize that algal decomposition in Hg-contaminated paddy fields can increase microbial MeHg production in soil and MeHg bioaccumulation in rice. The possible pathways would be involved with 1) the release of MeHg from decayed algal aggregate; 2) labile AOM-related increases in microbial MeHg production in soil; or 3) supply of LMW thiol-related MeHg translocation assistors. A simulative algal decomposition was performed in the micron-sized litterbags through pot and microcosm experiments, using Hg-contaminated paddy soils from the Wanshan Hg-mining area in China. Soil, algal, and rice samples were collected during the rice-tillering, grain-filling, and mature periods. Combined with chemical analysis, high-throughput sequencing, and multiple spectral techniques, the response characteristics of SDOM compositions and properties, microbial community, water-soluble cysteine, and total Hg (THg)/MeHg concentrations in the soil, rice or algae to algal decomposition were revealed. The results are expected to provide an understanding of the algal-involved regulation of the production and bioaccumulation of MeHg in Hg-contaminated paddy fields.

2 Materials and methods

2.1 Material characteristics

According to our previous investigation, filamentous algae such as *Ulothrix* sp. was the dominant species in the Wanshan Hg-mining paddy fields. Filamentous algae and rice seedlings (*Yixiangyou 2115*) were collected from a Hg-free paddy field in Huaxi district, Guizhou Province, China (26°23′45″N, 106°39′44″E). In this process, filamentous algae in the overlying water were collected using a nylon sieve (60 mesh, 250 μm aperture size), which was then repeatedly cleaned with ultrapure water to remove

impurities, including minerals and animals, in the laboratory. Filamentous algae have a larger size and may have inter-individual differences in Hg distribution. They were first vacuum-freeze-dried (−60 °C), and then ground through a nylon sieve (60 mesh) to achieve homogeneous algal powders. They had C:N ratios much lower than 25:1, conducive to microbial mineralization (Table S1) (Leite et al., 2017).

Topsoil (0−20 cm) was collected from Hg-contaminated paddy fields in the Wanshan Hg-mining area, Guizhou Province (27°33′51″N, 109°11′35″E). This is one of the largest Hg mines in the world, causing serious soil Hg contamination in the local paddy fields, up to 187.8 mg/kg (THg) (Yin et al., 2018a). Frequent algal blooms were observed in these paddy fields (Fig. S1). In this study, the selected soil had a THg concentration of 5.2 mg/kg (Table S1), which represented the Hg level in most rice fields in the Wanshan Hg-mining area, but far exceeded the safe value designated by the Chinese government (GB 15618-2018, 0.6 mg/kg, 6.5 < pH < 7.5). The soils were first air-dried before impurities, including plant residue, were removed. The soils were then ground and passed through a nylon sieve (60 mesh for the pot experiment or 100 mesh [147 μm aperture size] for the microcosm experiment). Rice seedlings with a similar growth status (e.g., height and weight) were selected for the pot experiment.

2.2 Experimental designs

A pot experiment was conducted in a plastic bucket (5 L), inside which 2 kg of Hg-contaminated soil was added (Fig. S1). Tap water with Hg level of less than 1.0 μg/L was then added to keep the overlying water at a depth of approximately 10 cm. After suspended particulate matter sedimentation for 24 h, the rice seedlings were transplanted manually. After entering the late tillering stage, 3.5 g ($A_{3.5}$), 7 g (A_7), and 15 g (A_{15}) of algal powders (dry weight) were added. To simulate DOM input from algal decomposition or decrease algal coarse particle input, algal powders were packed in the 1600 mesh nylon litterbags (approximately 9 μm aperture size). They floated in the overlying water around the rhizosphere without direct contact with the surface soil. According to previous investigations, the algal biomass (dry weight) in paddy fields in different regions fluctuated greatly; for example, it can range from 294 to 2315 kg/hm^2 (i.e., 0.03 to 0.23 kg/m^2) in a county of Hunan Province, China (Wu, 2021). In this study, the added biomass of algal powders (dry weight) ranged from 0.09 to 0.35 kg/m^2, near the scope of field observations from paddy fields (Wu, 2021; Wang et al., 2022). The algae-free treatment was set as the control. Three replications were conducted for each treatment and the control. Self-generating planktonic algae with fine size in the algal treatments and the control were weekly removed using a 200-mesh screen (75 μm aperture size) during the rice-growing period. This handling strategy may result in less impact on the Hg mass balance in the soil-water system as the small size of self-generating algae and huge soil Hg pool. The overlying water level was kept steady by a daily supplement of tap water.

To understand the role of decayed algal aggregates in regulating MeHg distribution in the soil-water system, a microcosm experiment was conducted in a 100 mL centrifuge tube (PTFE, Corning, USA) reactor (Fig. S1). Two treatments were established with and without Hg-contaminated soil substrate, i.e., "algae + water" (AW) and "algae + soil + water" (AWS). For the AW treatment, 60 mL ultrapure water was added to the reactor, and 0.2 g algal powders packaged in nylon litterbags (1600 mesh) were soaked beneath the water surface. For AWS, 40 g of soil was weighed in the bottom of the reactor before adding 60 mL of ultrapure water and 0.2 g of algal powders to the litterbags, which were suspended in the overlying water above the soil surface. All reactors were operated with three replications. Thereafter, they were incubated at a constant temperature of 26 °C in an artificial climate chamber (SPX-250BIII, Tianjin, China) for 25 days.

2.3 Sampling events and sample pretreatment

In the pot experiment, soil, rice tissues, and algal samples were collected. Soil samples were collected during the tillering, filling, and mature periods of rice growth. In every sampling event, multipoint soils in each pot were randomly collected from the subsoil layer (approximately 5 cm) around the root rhizosphere following a principle of minimum sampling mass and ultimately mixed into one soil sample. Soil samples were then divided into several subsamples: 1) Soil 1 was stored at −20 °C for water-soluble cysteine extraction; 2) Soil 2 was vacuum-freeze-dried at −60 °C and ground through a nylon sieve (100 mesh) for the determination of MeHg, THg, dissolved organic carbon (DOC) concentrations, and DOM properties (i.e., ultraviolet-visible, fluorescent, Fourier transform infrared (FTIR), and nuclear magnetic resonance (NMR) spectroscopies); and 3) Soil 3 was stored at −80 °C for microbial DNA extraction. Soil microbial communities were measured to understand the variation of potential Hg methylators driven by AOM input. At the mature period, rice tissues including husks and grains were collected, washed with ultrapure water, freeze-dried, and ground through a nylon sieve (100 mesh) to determine their THg and MeHg concentrations. Furthermore, algal debris in the litterbags was also collected to carry out treatments similar to those undertaken for the rice tissues.

In the microcosm experiment, algal debris in the litterbags was collected at 0, 1, 3, 6, 10, 15, 20, and 25 d, gently washed with ultrapure water, vacuum-freeze-dried,

and ground through a nylon sieve (100 mesh) to determine mass loss, as well as THg and MeHg concentrations.

2.4 Chemical and spectral analysis

The THg concentrations in plant tissues (rice and algae) were determined using a DMA-80 Direct Hg Analyzer (Milestone, Italy) according to USEPA Method 7473. The protocols for soil THg, soil MeHg, and plant MeHg have been reported elsewhere in detail (Liang et al., 1996; Zhang et al., 2010). Briefly, THg concentrations in soils were determined by first digesting material in a water bath (95 °C) using a fresh mixture of HCl and HNO_3 (3:1, *v/v*) and then subjecting samples to a sequential procedure including BrCl oxidation, $SnCl_2$ reduction, gold adsorption, thermal adsorption, and detection by cold vapor atomic fluorescence spectrometry (CVAFS, Brooks Rand Co., America). The MeHg concentrations in plant tissues were determined by first digesting material in KOH-methanol solution (250 mg/L) and then subjecting them to a sequential procedure including extraction (CH_2Cl_2), back-extraction (ultrapure water), ethylation ($NaBEt_4$), Tenax adsorption, thermal adsorption, gas chromatographic separation, and detection by CVAFS. In contrast, the MeHg concentrations in soil samples were measured by first extraction using $CuSO_4$ (2 mol/L) and HNO_3 (25%, *v/v*) and then following a similar procedure as for the plant material.

To measure the water-soluble cysteine concentration, DOC concentration, and DOM compositions and properties (i.e., ultraviolet-visible, fluorescent, FTIR, and ^{13}C NMR spectroscopies), the soil to water ratio (1:10) was conducted in 50 mL centrifuge tubes, which were oscillated in a reciprocating manner (180 r/min) at 25 °C for 12 h. After centrifugation at 4000 r/min for 30 min, the supernatants were passed through 0.45 μm filters (PTFE) to collect the SDOM filtrates, which were then subjected to chemical or spectral analysis. Related analytical methods and spectral implications can be accessed in the supporting information (Text S1**).** In this study, ultraviolet-visible spectroscopy was used to reveal the aromaticity and molecular weight of SDOM, and fluorescent spectroscopy combined with parallel factor analysis (PARAFAC) was used to estimate the humification degree and protein- and humic-like fractions in the SDOM (Nebbioso and Piccolo, 2013). Several spectral implications are briefly listed here. The a(355) represents the colored DOM (CDOM), which is part of the DOM that absorbs blue light, indicating the physical and chemical conditions in freshwater (Helms et al., 2008). $SUVA_{254}$ reflects the DOM aromaticity, the higher the $SUVA_{254}$ value, the higher the DOM aromaticity (Weishaar et al., 2003). The spectral slope ratio (S_R) reflects the DOM molecular weight, the lower the S_R value, the higher DOM molecular weight (Helms et al., 2008). Lastly, the Humification index (HIX) reflects the DOM humification level, with higher HIX values indicating higher humification levels (Ohno, 2002). FTIR spectroscopy was used to provide information related to the chemical bond and functional groups in SDOM (Ohno, 2002), and ^{13}C NMR spectroscopy was used to understand the molecular composition of SDOM, such as C-containing functional groups (Huang et al., 2021).

Microbial communities in soil from the pot experiments were analyzed using a high-throughput sequencing technique. Total DNA was extracted from approximately 0.5 g of fresh soil with a FastDNA™ SPIN Kit (MP Biomedicals, Illkirch, France). The V4 regions of the 16S rRNA genome were amplified using the primer pair 515F (5′-GTGYCAGCMGCCGCGGTAA-3′) and 806R (5′-GGACTACNVGGGTWTCTAAT-3′). The detailed analytical procedures, including PCR, can be accessed in the supporting information (TEXT S2**)**, and the PCR products were sequenced by Majorbio Biopharm Technology Co. Ltd. (Shanghai, China). Here, microbial communities were divided into two categories: 1) Hg methylators, defined as the taxa that carry the gene cluster *hgc*AB, and 2) putative non-Hg methylators, defined as the taxa that do not carry *hgc*AB. Recent studies found that non-Hg methylators also play important roles in affecting Hg methylation of Hg methylators through the trophic interactions between them (Liu et al., 2019b; Xu et al., 2019). The shifts of non-Hg methylators could also be one of the key factors controlling soil's MeHg production in the paddy soils (Liu et al. 2019b).

2.5 Quality control and quality assurance

Quality assurance and quality control for the chemical analysis were conducted using duplicate analyses, matrix spikes, and/or standard reference materials (SRMs). For THg in soils and plants, GSB-11 (citrus leaf, GBW10020), GSS-5 (soil, GBW07405), and a standard solution of $HgCl_2$ (liquid, 100 μg/mL, Nonferrous Research Institute, Beijing, China) were used as SRMs. For MeHg in soils and plants, ERM-CC580 (estuarine sediment), TORT-3 (lobster hepatopancreas), and a standard solution of MeHg (liquid, 1 μg/mL, National Institute of Metrology, Beijing, China) were used as SRMs. The THg and MeHg recoveries for the SRMs and matrix spikes were 85%–110%, and the relative standard deviations (RSDs) for all duplicate samples were < 15%. For microbial community analysis, the average coverage of all samples exceeded 0.97, indicating that the identified sequences were representative of the actual community in the sample.

2.6 Statistical analyses

Statistical analyses of the data were performed using SPSS 25.0 and Origin 2021 software. One-way analysis

of variance (ANOVA) with Tukey's multiple comparison tests was used to compare the statistical significance between the control, $A_{3.5}$, A_7, and A_{15}. MATLAB 2016b, combined with the EfC platform developed by China University of Geosciences and Peking University (v1.0), was used to perform parallel factor analysis (PARAFAC). A detailed method can be accessed in the TEXT S3, supporting information.

3 Results

3.1 Soil DOM properties

3.1.1 DOC and water-soluble cysteine

During the tillering period, dissolved organic carbon (DOC) concentration in the soil with the addition of the highest algal biomass (A_{15}) was significantly higher than the control (0.58 ± 0.02 mg/g *vs.* 0.74 ± 0.03 mg/g, $P < 0.01$), whereas it was not significant at low algal dosages ($A_{3.5}$ or A_7) or during the other rice-growing periods (Fig. 1(a)). Considering the rice-growing periods, the soil DOC concentration in the control declined slightly over time ($P > 0.05$), while it decreased sharply reduced in the algal treatments ($P < 0.05$). In particular, the reduction percentage in A_{15} (28.46%) was higher than that in $A_{3.5}$ (20.52%) and A_7 (14.34%); the more the algal biomass added, the higher consumption of the SDOM.

Compared to the control (31.42 ± 2.05 μg/g), the water-soluble cysteine concentrations in the soil were significantly higher in A_7 (40.00 ± 1.06 μg/g) and A_{15} (42.99 ± 0.32 μg/g) during the rice-tillering period ($P < 0.05$, Fig. 1(b)). Similar results were also found during the grain-filling periods. The maximum percentage increases during the rice-tillering and grain-filling periods were 36.8% and 23.9%, respectively. However, the water-soluble cysteine concentrations in the soil were slightly higher in $A_{3.5}$ and A_{15} than those in the control during the mature period ($P > 0.05$). During the different rice-growing periods, the water-soluble cysteine concentrations in soil in the control followed an order of grain-filling period (44.87 ± 4.14 μg/g) > rice-tillering period (31.42 ± 2.05 μg/g) and mature period (33.55 ± 1.92 μg/g). In contrast, algal treatments with different biomasses also followed a similar pattern that water-soluble cysteine concentrations in soil in the grain-filling period (49.66 ± 5.38 μg/g) were higher than those in the rice-tillering period (38.26 ± 5.81 μg/g) and the mature period (36.36 ± 4.80 μg/g).

3.1.2 Spectral signatures

Compared to the control, soil CDOM concentrations and S_R value were significantly elevated by adding different algal dosages during the rice-tillering period ($P < 0.05$), but were not significantly different during the other rice-growing periods ($P > 0.05$) (Figs. 2(a)–2(b)). Soil CDOM concentrations increased from 45.12 ± 3.86 to 48.73 ± 1.75 m^{-1}, driven by the lowest to highest algal biomass

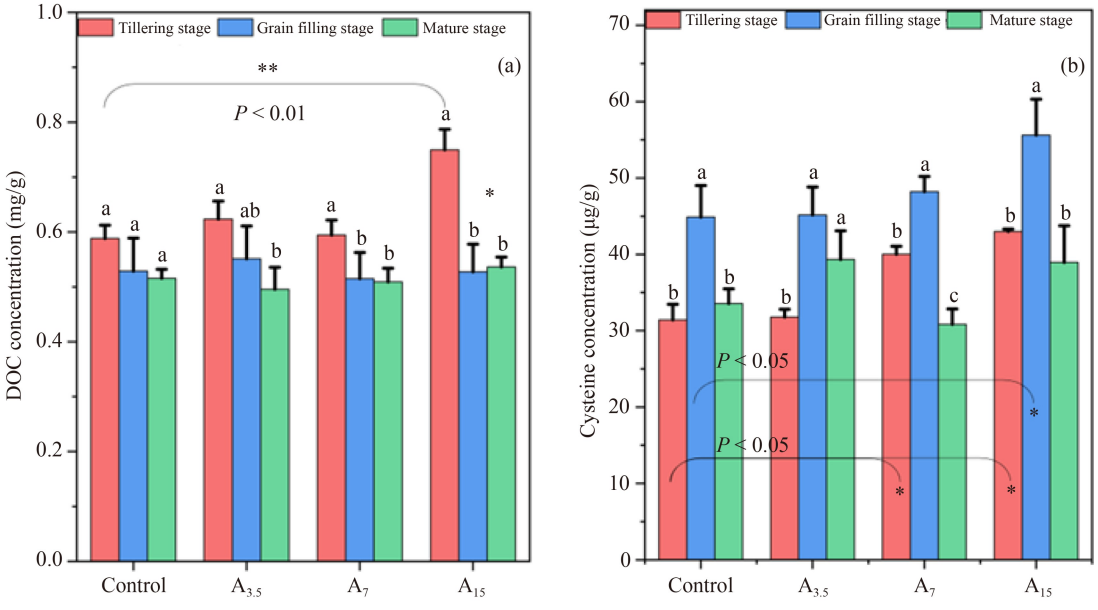


Fig. 1 DOC (a) and water-soluble cysteine (b) concentrations in Hg-contaminated soils regulated by algal decomposition. Data are represented by average ± standard deviation (AVE ± SD). Different lowercase letters indicate significant differences between different rice growing periods at the same treatment at 0.05 level. "**" and "*" represent the significant differences between algal treatments and the control at 0.01 and 0.05 levels at the same rice growing period, respectively. ($A_{3.5}$, A_7, and A_{15} = algal treatments with 3.5, 7, and 15 g biomass, respectively).

during the rice-tillering period, while the S_R values increased significantly from 0.98 ± 0.02 to 1.12 ± 0.08 ($P < 0.05$). The soil CDOM concentrations in the algal treatments were significantly reduced during the rice-growing periods ($P < 0.05$), which was different from the results observed in the control (Fig. 2(a)). The algal treatments had relatively higher S_R values during the grain-filling period than other periods, and were also different from those of the control (Fig. 2(b)). Moreover, the humification level (HIX) and aromaticity ($SUVA_{254}$) of SDOM in the algal treatments had lower values during the grain-filling period than during the other periods (Figs. 2(c)–2(d)).

As shown in TEXT S3, the PARAFAC method picked out two fluorescent components, i.e., humic-like components (C1, C2, and C3) and a protein-like component (C4) (Fig. 3(c)). Here, the maximum fluorescence intensity (F_{max}) of the SDOM components was calculated, indicating their relative abundance (Figs. 3(a)–3(b)). The protein-like component C4 decreased gradually by 10% from the lowest to highest addition of algal biomass ($P < 0.05$), while the humic-like components (C1, C2, and C3) had less variation ($P > 0.05$) (Fig. 3(a)). Considering the rice-growing periods, the fractions of C1, C2, C3, and C4 in the control and algal treatments decreased significantly over time ($P < 0.05$), except for C2 in the algal treatment ($P > 0.05$) (Fig. 3(b)). It should be noted that the lower average reduction percentage of the component C4 in the algal treatment (9.70%) differed considerably from the control (16.76%).

FTIR spectroscopy of SDOM showed that algal decomposition altered the intensities of the functional groups (Fig. S2). An antisymmetric stretching vibration of CO_2 at 2400–2200 cm^{-1} was only found during the rice-tillering period in the algal treatments, but not during the other periods. Moreover, the absorption peak at 869 cm^{-1} (aromatic carbon) had no variation during the rice-tillering period, but was elevated during the other periods. ^{13}C NMR spectroscopy showed that the average carboxyl carbon fraction in SDOM increased from 1.43% to 5.92% in the control during the rice-growing periods, while it increased from 10.25% to 17.19% in the algal treatments. In other words, the average carboxyl carbon fraction in the SDOM of the algal treatments increased greatly and was increased by 2.90–7.17 times higher than that in the control.

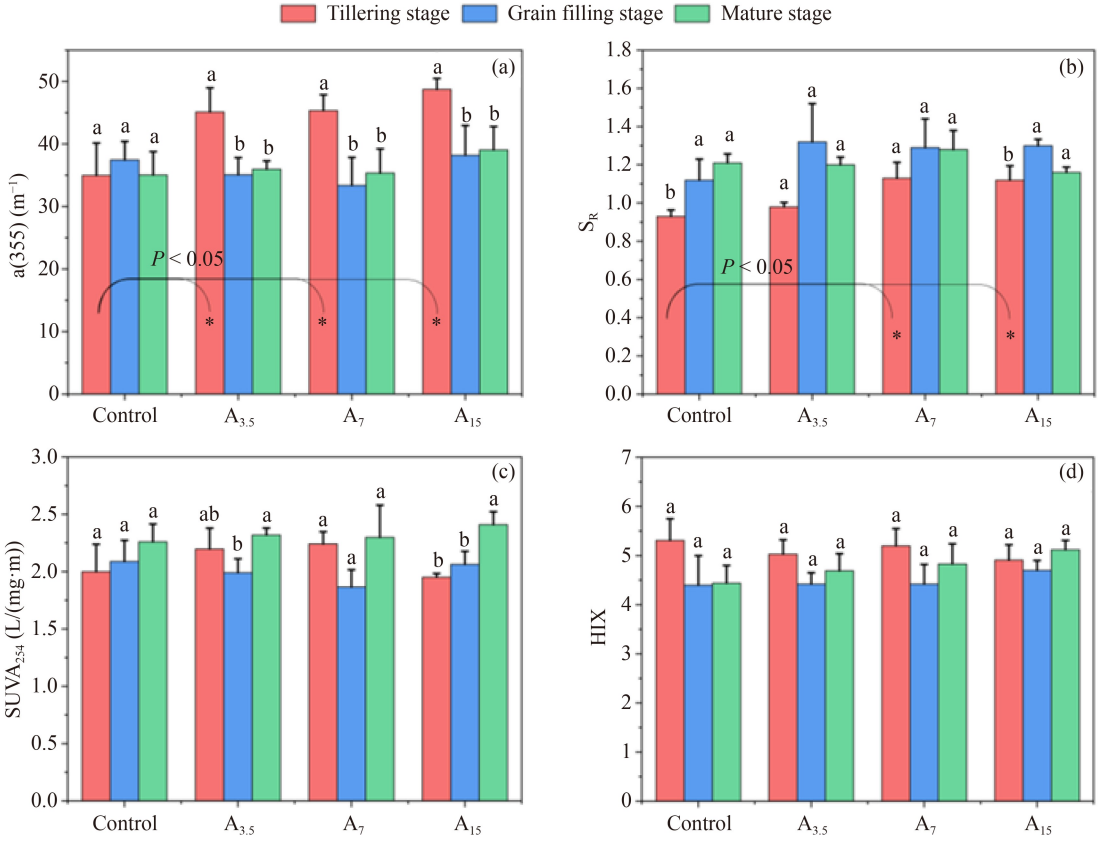


Fig. 2 Optical properties of soil DOM including a(355) (a), S_R (b), $SUVA_{254}$ (c) and HIX (d) regulated by algal decomposition. Data are represented by AVE ± SD. Different lowercase letters indicate significant differences between different rice growing periods at the same treatment at 0.05 level. "*" represents the significant differences between algal treatments and the control at 0.05 level at the same rice growing period. ($A_{3.5}$, A_7, and A_{15} = algal treatments with 3.5, 7, and 15 g biomass, respectively).

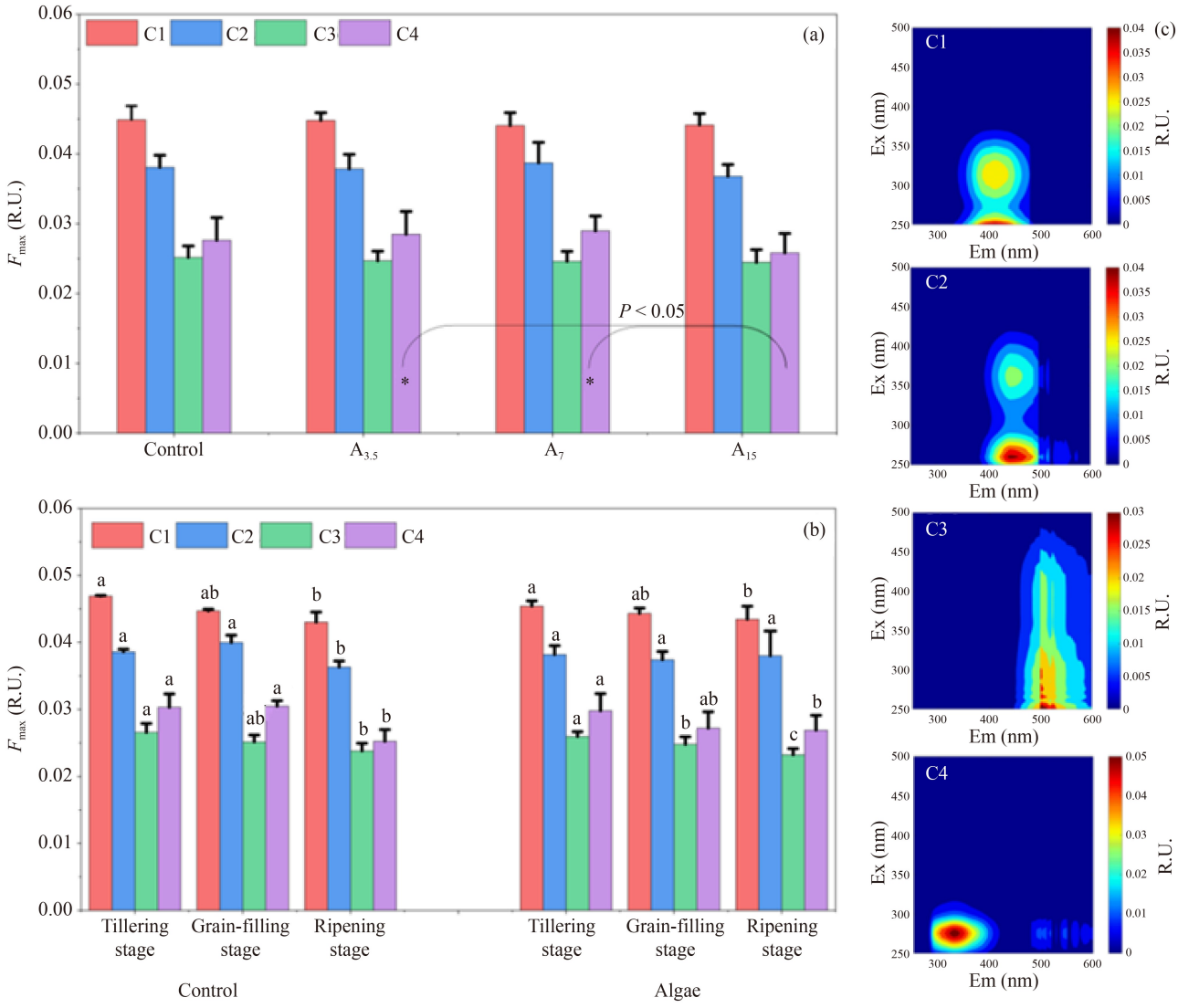


Fig. 3 Fluorescent components in soil DOM regulated by algal decomposition (a–b), and their 3D-EEMs (c). C1, C2 and C3 represent humic-like components, and C4 represents protein-like components. Data are represented by AVE ± SD. Different lowercase letters indicate significant differences between different rice growing periods in the control or algal treatment at 0.05 level, and "*" represents the significant differences between algal treatments at 0.05 level. ($A_{3.5}$, A_7, and A_{15} = algal treatments with 3.5, 7, and 15 g biomass, respectively).

3.2 Bacterial community

Algal decomposition increased the microbial species to 1.18 times higher than the control during the rice-tillering period, but not during other periods (Fig. S3). This result was further verified by the Ace, Chao, and Shannon indices, which are used to reflect the richness and diversity of the microbial species (Table S4). At the phylum level, we found that non-Hg methylator *Cyanobacteria* (Lázaro et al., 2019) accounted for 23.03% of the total microbial abundance during the grain-filling period, which was significantly higher than that during the rice-tillering periods (3.60%) and mature period (1.80%). Based on the Hg-MATE database (v1.01142021) (Gionfriddo et al., 2021), several potential MeHg regulators now known to be related to MeHg production were identified at the genus level, including Hg methylators (*Anaerolineaceae*, *Bacteroidetes*, *Anaerolinea*, *Desulfocapsaceae*, and *Geobacteraceae*) and a non-Hg methylator (*Cyanobacteria*) (Fig. 4(b)). Compared to the control, algal decomposition increased the total abundance of potential Hg methylators in soils by 11.47% and 150.82% during the grain-filling and mature periods, respectively. Additionally, it also greatly increased the abundance of potential non-Hg methylators in soils by 721.42% during the grain-filling period. However, algal decomposition resulted in no significant changes in Hg methylators and non-Hg methylators in soils at the genus level during the rice-tillering period compared to the control.

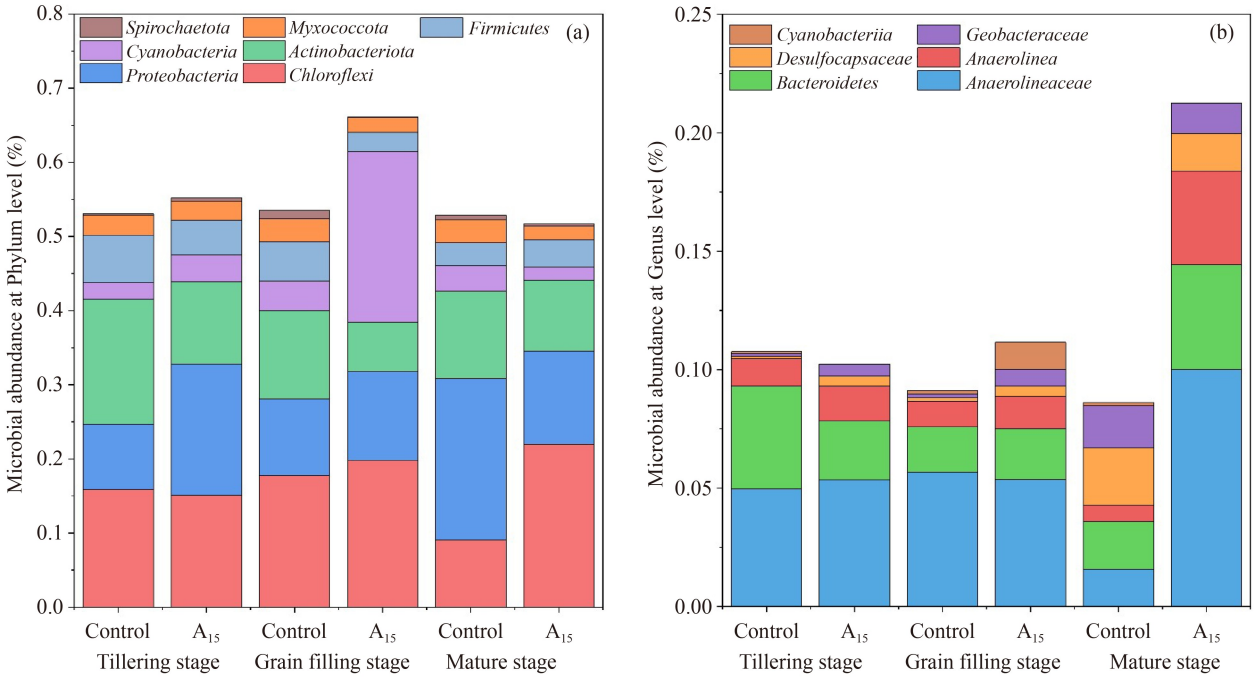


Fig. 4 Abundance of potential MeHg regulators including Hg methylators and non-Hg methylators regulated by algal decomposition at Phylum (a) and Genus (b) levels. (A_{15} = algal treatments with 15 g biomass).

3.3 Hg in algal particles

The initial absolute THg concentrations in algal powders in the $A_{3.5}$ and A_{15} were 158.34 and 678.60 ng, which increased to 189.47 ± 4.49 and 761.47 ± 133.15 ng during the mature period, respectively (Fig. 5(a)). In addition, the initial absolute MeHg concentrations in the A_7 and A_{15} were 15.96 and 34.20 ng, which also increased to 16.97 ± 2.22 and 71.72 ± 3.23 ng during the mature period, respectively.

The microcosm experiment revealed a quick mass loss of algal powders within 15 d, higher than 65%, which was consistent with the result in the pot experiment (approximately 70%) (Fig. 5(b)). Absolute THg concentrations in the decayed algal powders decreased significantly and then increased quickly over time in the AWS ($P < 0.05$), whereas they followed a steady decline in the AW ($P < 0.05$) (Fig. 5(c)). Moreover, absolute MeHg concentrations in the decayed algal particles decreased greatly from 0.63 ± 0.02 ng to 0.09 ± 0.01 ng within 10 d, and then increased slightly to 0.22 ± 0.01 ng in 25 d in the AW ($P < 0.05$). Additionally, they also decreased from 0.67 ± 0.01 ng to 0.17 ± 0.02 ng within 10 d, followed by a substantial increase to 0.45 ± 0.06 ng by 25 d in the AWS ($P < 0.05$) (Fig. 5(d)). The MeHg/THg ratio (%MeHg) in the decayed algal particles gradually decreased from 6.69 ± 0.24% to 2.47 ± 0.12% within 10 d, and increased greatly to 6.73 ± 0.45% by 25 d in the AW. In contrast, it first decreased from 8.27 ± 0.67% to 3.38 ± 0.42% and then increased to 4.83 ± 0.54% in the AWS.

3.4 Hg in the soil-rice system

Compared to the control, the addition of low to high algal biomass resulted in pronounced increases in soil MeHg concentrations by 25.6%–80.2% and 12.6%–66.1% during the tillering and grain-filling periods, respectively ($P < 0.05$) (Fig. 6(a)). In comparison, soil MeHg concentrations during the mature period only increased significantly when adding the highest algal biomass (13.37 ± 0.86 ng/g *vs.* 20.36 ± 1.18 ng/g) ($P < 0.05$). Among the rice-growing periods, the grain-filling period had significantly higher soil MeHg concentrations than the others ($P < 0.05$).

Compared to the control (80.06 ± 3.23 ng/g), the THg contents in rice grains were increased significantly in A_7 (90.07 ± 1.26 ng/g) and A_{15} (130.39 ± 7.14 ng/g) ($P < 0.05$, Fig. 6(b)). Similarly, compared to control (45.93 ± 2.56 ng/g), MeHg concentrations in rice grains were slightly increased in A_7 (49.11 ± 1.46 ng/g, $P > 0.05$) but greatly elevated in A_{15} (79.14 ± 0.85 ng/g, $P < 0.05$) (Fig. 6(c)). The bioaccumulation factor (BAF) of MeHg (grain MeHg/soil MeHg ratio) was 3.44 ± 0.12 in the control, which increased from 3.28 ± 0.26 ($A_{3.5}$) to 4.01 ± 0.21 (A_{15}) (Fig. 6(d)). The algal decomposition greatly increased the MeHg concentrations and BAF in rice grains, by 72.30% and 16.77%, respectively. THg and MeHg in the rice grains and husks showed similar response characteristics to the algal-derived OM input (Fig. 6). However, algal decomposition greatly accelerated the Hg accumulation in rice grains compared with

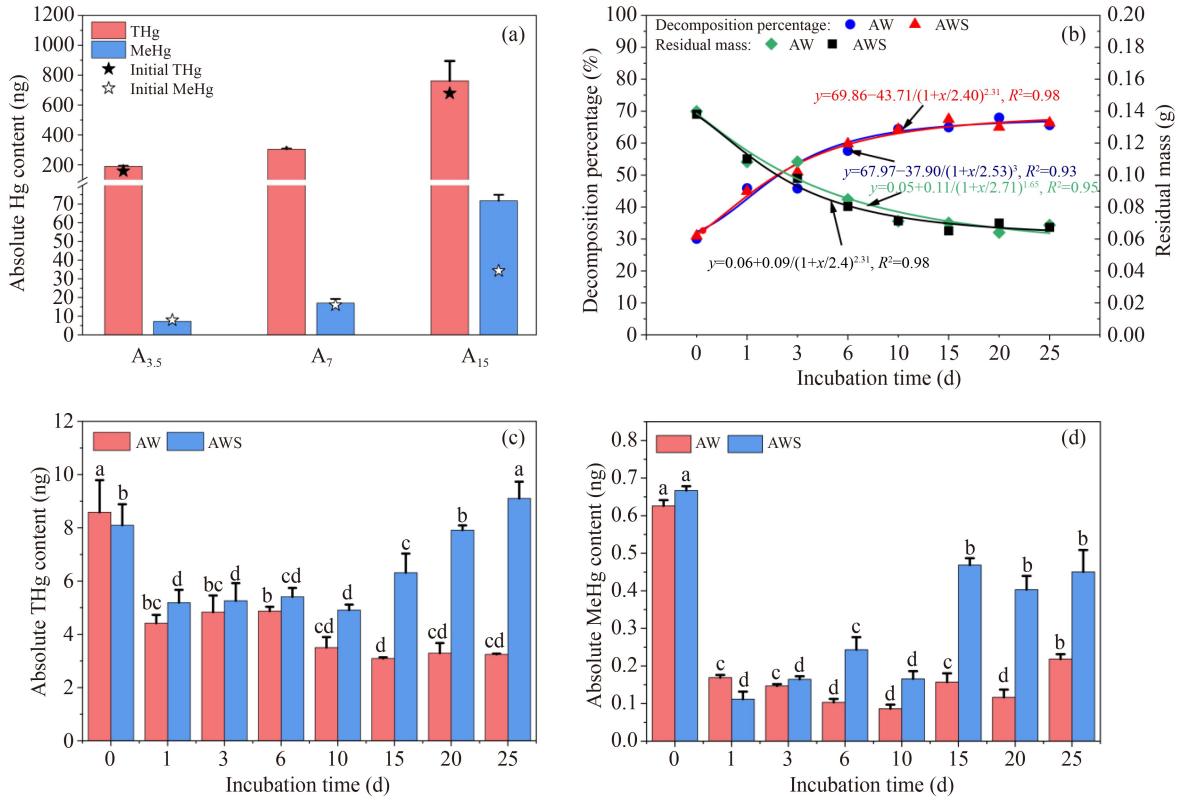


Fig. 5 Absolute THg and MeHg contents in the decayed algal particles in the pot experiment (a). Decomposition percentage and residual mass of algal powders in the microcosm experiment (b). Absolute THg (c) and MeHg contents (d) in the algal particles in the microcosm experiment. Data are represented by AVE ± SD in the graphs a, c and d. Different lowercase letters indicate significant differences between different incubation time at 0.05 level. ("AW" = "algae + water"; "AWS" = "algae + water + soil"; $A_{3.5}$, A_7, and A_{15} = algal treatments with 3.5, 7, and 15 g biomass, respectively).

that in husks, with the largest increases (the ratio of ΔHg in grains to ΔHg in husks; Δ represents the Hg increases between algal treatments and control) being 589% and 346% for THg and MeHg, respectively.

4 Discussion

4.1 Responses of soil DOM properties to algal-derived OM input

Significantly elevated DOC concentrations in soil mainly occurred during the rice-tillering period, especially with the highest addition of algal biomass (Fig. 1(a)). Combined with the results of the decomposition rate of algal powders (Fig. 5(b)) and the CO_2 peak variation of SDOM (Fig. S2), we speculate that algal powders were subjected to quick microbial consumption in a relatively short time. The lower C:N ratio (Table S1), combined with the abundant bioavailable fractions such as proteins, oligosaccharides, and organic acids in algae, would favor the preferential biodegradation of AOM (Bai et al., 2017; Yang et al., 2022).

Water-soluble cysteine concentrations in soils were significantly higher during the grain-filling period than during other rice-growing periods in the control. They were greatly elevated with the addition of algal powders during the grain-filling and tillering periods (Fig. 1(b)). Many studies have found that AOM is rich in LMW thiol compounds, such as cysteine (Bai et al., 2017; Bouchet et al., 2018; Lei et al., 2021), which may be responsible for the elevated cysteine concentrations in soil, which are driven by algal decomposition. Moreover, root metabolism is usually involved in the release of LMW organic acids (Yin et al., 2018b). Rice roots may suffer from vigorous metabolism during the grain-filling period due to the higher root system and nutritional requirements (Zhu et al., 2020), theoretically explaining the higher soil water-soluble cysteine concentrations in the control during the grain-filling period (Fig. 1(b)).

Algal decomposition induced significant increases in soil CDOM concentrations at the beginning of rice growth; thereafter they were greatly reduced (Fig. 2(a)). Algal decomposition releases much fresh intracellular DOM at the water-soil interface, increasing soil CDOM. However, it is rich in labile organic fractions and is readily consumed by bacteria (Hanamachi et al., 2008; Hulatt et al., 2009). Moreover, algal-derived OM input

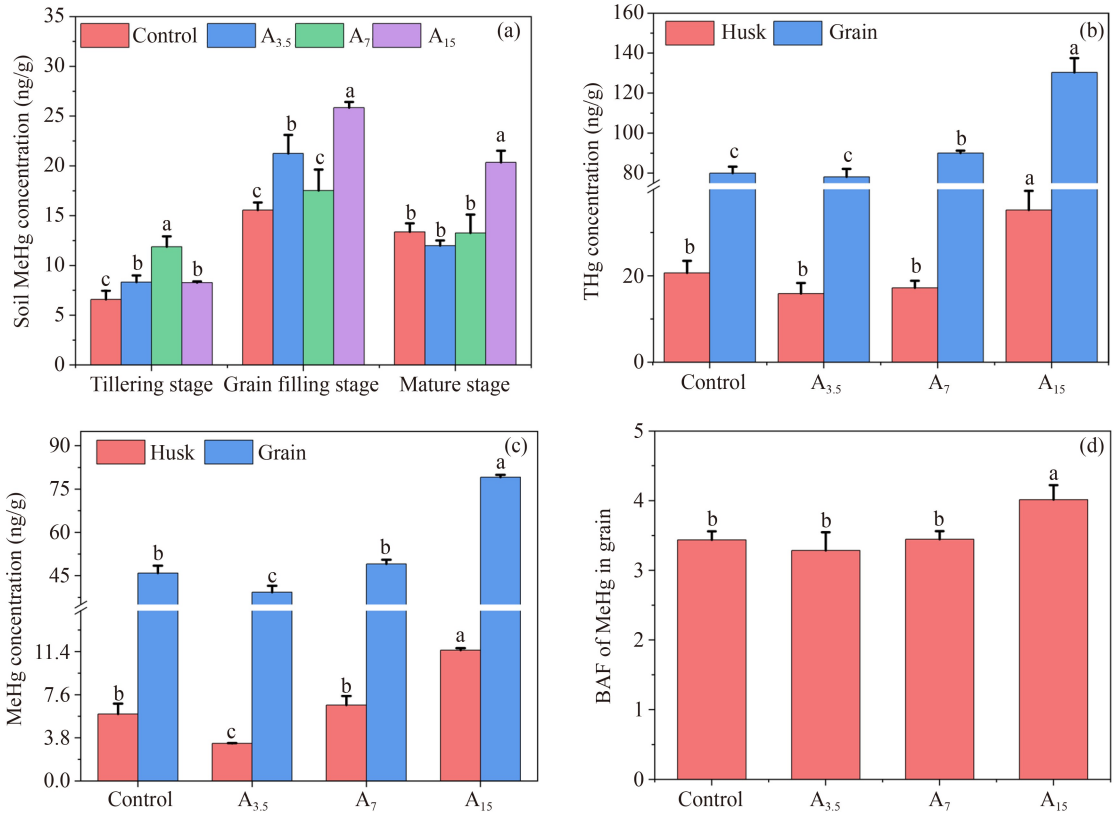


Fig. 6 MeHg concentrations in soil (a), THg concentrations in husks and grains (b), MeHg concentrations in husks and grains (c) and bioaccumulation factor (BAF) of MeHg in grains (d) regulated by algal decomposition. Data are represented by AVE ± SD. Different lowercase letters indicate significant differences between different algal treatments and the control at 0.05 level. ($A_{3.5}$, A_7, and A_{15} = algal treatments with 3.5, 7, and 15 g biomass, respectively).

lowered the molecular weight of SDOM during the rice-tillering and grain-filling periods (Fig. 2(b)) and also lowered the aromaticity and humification levels of SDOM during the grain-filling period (Figs. 2(c)–2(d)). This result indicates that AOM results in a decrease in molecular aggregation and an increase in LMW fractions of SDOM, possibly due to the higher input of fresh AOM. This finding was partially supported by the elevated levels of water-soluble cysteine in the soil as well as the significant negative relationship between cysteine and $SUVA_{254}$ ($P < 0.05$, Fig. 7(a)).

Algal decomposition could release much AOM rich in the form of protein-like components, such as amino acids (Hanamachi et al., 2008; Hulatt et al., 2009). However, a significant decrease in the protein-like component in SDOM by adding low to high algal biomass indicates that increasing AOM input can accelerate the microbial consumption of algal-derived protein materials (Fig. 3(a)). It may also reflect the elevated bacterial activity or abundance in response to AOM addition (Fig. S3 and Fig. 4). Nevertheless, compared with the control, algal treatment lowered the decreasing percentage of the protein-like components in SDOM over time (Fig. 3(b)). Here, we speculate that the protein supplements from microbial biosynthesis and AOM may delay the decrease in the protein-like component in SDOM.

Overall, sedimentation of algal-derived OM indeed changes the SDOM properties and compositions, and it tends to shape an SDOM medium with low molecular weight, aromaticity, and humification, as well as a series of LMW organic ligands.

4.2 Algal-derived OM increases the soil MeHg accumulation

Paddy soil MeHg is mainly sourced from microbial *in situ* synthesis and exogenous input (e.g., fertilizers, sewage, and pesticides), whereas the roles of algal blooms were traditionally overlooked. The Hg variations in the algal aggregation in the pot and microcosm experiments indicate that decayed algal powder is both a source and sink of ambient Hg and synchronously experienced *in situ* Hg methylation (Fig. 5). The inorganic Hg released from soil or algae was adsorbed by the decayed algal aggregate, combined with the adhered labile AOM, minerals, and Hg methylators, greatly increasing the microbial MeHg production (Gascón Díez et al., 2016; Bouchet et al., 2018; Lei et al., 2019; Zhao

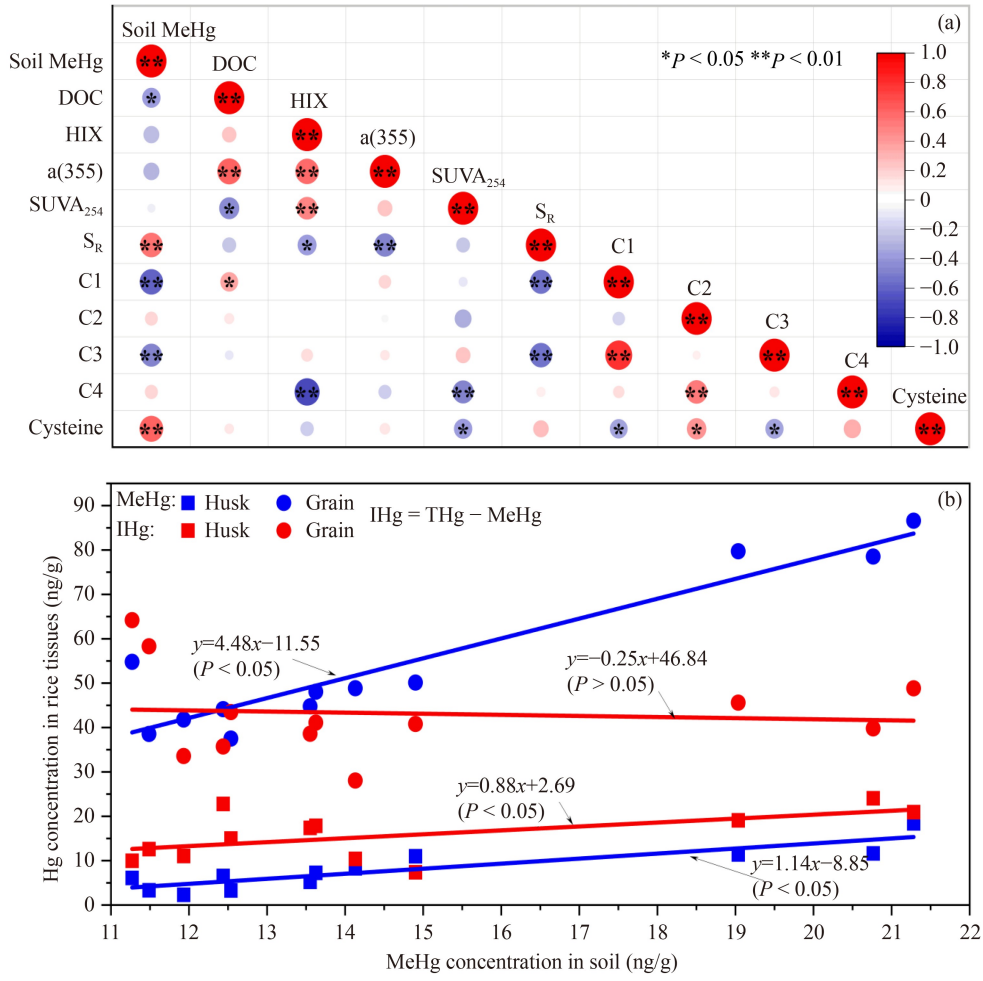


Fig. 7 Pearson's correlation matrix of soil MeHg concentration with soil DOM properties (DOC concentration, HIX, a(355), $SUVA_{254}$, S_R, C1, C2, C3 and C4) and soil cysteine concentrations (a); Linear fitting between soil MeHg concentration and inorganic Hg (IHg = THg minus MeHg), or MeHg concentration in rice tissues (b).

et al., 2021). These synthesized MeHg may enter the water-soil interface coupled with algal decomposition. Moreover, algal decomposition also released considerable inorganic Hg for Hg methylators in soil (Fig. 5(a)).

Many studies found that DOM usually had a two-sided role (i.e., promotion or inhibition) in regulating microbial MeHg production, which is controlled by its inherent properties (He et al., 2019). Although soil DOC levels were highest during the rice-tillering period (Fig. 1(a)), we found that elevated MeHg production mainly occurred during the grain-filling period (Fig. 6(a)). Furthermore, higher soil DOC concentrations tended to have a potentially inhibiting effect, as revealed by the significantly negative relationship between soil MeHg and DOC ($P < 0.05$, Fig. 7(a)). This result suggests that elevated MeHg production in algae-amended soils cannot be explained by the algal-driven increase in SDOM concentrations. We, therefore, conclude that elevated MeHg production in the soil is related to the AOM-driven variations of SDOM molecular properties and microflora.

First, algal decomposition resulted in a significantly higher water-soluble cysteine concentration in soil than the control (Fig. 1(b)), and it had significant positive relationships with soil MeHg concentration ($P < 0.01$, Fig. 7(a)), indicating that elevated water-soluble cysteine tends to greatly promote soil Hg methylation. Algal-derived OM is rich in LMW thiol-containing compounds such as cysteine (Bai et al., 2017; Bouchet et al., 2018; Lei et al., 2021), which can compete for soil-bound Hg^{2+} to form a cysteine-Hg complex, increasing the Hg bioavailability (Liem-Nguyen et al., 2017). The formation of the cysteine-Hg complex can also facilitate the rate of the Hg^{2+} transformation reaction by promoting both the microbial uptake of inorganic Hg and the enzymatic formation of MeHg (Schaefer and Morel, 2009; Liem-Nguyen et al., 2017). In addition, the water-soluble cysteine is also an important labile carbon source for Hg-methylating bacteria, other than being an organic ligand donator. The increase in cysteine-related microbial activity or abundance may also be one of the most important contributors to the elevated MeHg production.

Second, algal decomposition resulted in a significantly

higher S_R value of SDOM than the control (Fig. 2(b)), and S_R value had a significantly positive relationship with soil MeHg concentration ($P < 0.01$, Fig. 7(a)). This result suggests that AOM input shapes an LMW SDOM medium, greatly promoting the higher MeHg production in the soil. A previous study showed that AOM was dominated by LMW organic components, with 81% having a molecular weight lower than 1 kDa (Lei et al., 2019). In addition, algal decomposition favored the longer retention time of protein-like fractions in SDOM (Fig. 3(b)). Furthermore, it also resulted in a considerable accumulation of carboxyl carbon fractions in SDOM, as indicated by the ^{13}C NMR results. These labile AOM components were more bioavailable, entering sediment and fueling the Hg-methylating bacteria to promote MeHg production (Bravo et al., 2017; Herrero Ortega et al., 2018; Lei et al., 2019). Especially, MeHg concentrations in algal-amended soils were significantly higher during the grain-filling period than during other periods. It may be also related to the labile SDOM characteristics, as revealed by the higher cysteine concentrations (Fig. 1(b)) but lower SDOM aromaticity (Fig. 2(c)) and humification (Fig. 2(d)) during the grain-filling period.

Third, we further revealed that algal decomposition greatly fueled several potential Hg methylators and non-Hg methylators in soil (Fig. 4). The contributions of Hg methylators are inevitable. Moreover, these non-Hg methylators also played significant roles in promoting MeHg production by impacting Hg bioavailability or providing a growth substrate (e.g., organic acids) for the Hg methylators (Liu et al., 2019b; Xu et al., 2019). Especially, non-Hg methylator *Cyanobacteria* was greatly fueled during the grain-filling period, matching the higher MeHg in soil. In many studies, *Cyanobacteria* were found to be an important MeHg regulator responsible for elevated MeHg production in wetland periphyton by providing habitat and growth substrates for the Hg methylators (Lázaro et al., 2013; Lázaro et al., 2019). Its blooms, driven by AOM input in Hg-contaminated paddy soils, may fuel the Hg methylators to enhance the MeHg production in soils.

4.3 Algal-derived OM increases MeHg bioaccumulation in rice

Algal decomposition at the soil-water interface greatly increased the MeHg bioaccumulation in rice grains, especially with the higher addition of algal biomass (Fig. 6). Here, we speculate that algal-boosted MeHg bioaccumulation in rice grains involves two main mechanisms.

First, the significantly positive relationship between rice MeHg and soil MeHg corresponded with prevailing findings that rice MeHg is mainly derived from soil (Fig. 7(b)) (Zhang et al., 2010; Meng et al., 2011). Algal-derived LMW OM fractions entered the soil matrix, fueling MeHg regulators or providing bioavailable Hg species, and thus promoting MeHg production. These MeHg could be either directly adsorbed by rice roots and accumulated in rice grains or bound by elevated water-soluble cysteine to form a cysteine-MeHg complex, which is readily absorbed and translocated in the root-shoot-grain system (Hao et al., 2022). As mentioned before, the higher MeHg production powered by elevated labile SDOM fractions mainly occurred during the grain-filling stage. Previous studies found that soil had elevated MeHg production during the rice-filling period, while rice had the greatest MeHg uptake before the rice-flowering period (Meng et al., 2011; Strickman and Mitchell, 2017; Guo et al., 2021). We thus believed that MeHg accumulation in soil elevated by algal decomposition during the grain-filling stage may greatly regulate the MeHg bioaccumulation in rice, which is worthy of further investigation.

Second, the MeHg and %MeHg variations in algal debris in the pot and microcosm experiments suggested that algal particles suffered from significant Hg release and in situ MeHg production, although they may also have a high capacity to adsorb MeHg from water, possibly due to the high thiol-containing ligands (Figs. 6(a), 6(c), and 6(d)). In this process, much LMW AOM-bound MeHg (e.g., the cysteine-MeHg complex) could enter the soil and be then accumulated by rice. Thus, decayed algal particles at the soil-water interface would be another possible source of MeHg in rice grains, which requires further study.

5 Environmental implications and notes

Algal blooms in the Hg-contaminated paddy fields are prevalent with considerable biomass, acting as an important regulator of soil carbon turnover (Wang et al., 2022). Although algal growth in the paddy fields plays a positive impact on carbon dioxide fixation, it also increased soil organic carbon availability (Wang et al., 2022). Many studies had observed that algal cycling in the lakes and wetlands was an important regulator involved with Hg methylation and MeHg bioaccumulation in the aquatic organism (Leclerc et al., 2015; Lei et al., 2019; Lei et al., 2021; Xiang et al., 2021). Our findings further emphasized that algal decomposition accompanied by rice growth in Hg-contaminated paddy fields was also an important contributor to the increase in production, bioaccumulation, and environmental risk of MeHg. Here, we suggest that algal decomposition is a non-negligible factor regulating the environmental behavior in the Hg-contaminated paddy fields. Feasible management strategies such as algal removal and rational fertilization are needed to be considered. However, less information can be accessed regarding the impacts of these management strategies, and further study is needed in the future.

Additionally, it should be emphasized that it was difficult to simulate a real field environment in the pot experiment. In the Hg-contaminated paddy fields, filamentous algae are dominated, while their distribution is greatly dynamic. In this study, we mainly paid attention to the crucial impacts of algal decomposition, which could contribute to much more AOM in soil-water systems. In this process, AOM supply efficiency may be affected by algal quality such as particle size and biomass, although environmental factors, such as temperature and microbial activity, are also important. However, the intricate spatial structure of filamentous algae may result in a great heterogeneity in their Hg levels and carbon distribution, causing a complex inter-individual difference in their decomposition characteristics and DOC release. The algal powders rather than filamentous algal aggregate were soaked in this study to avoid this question, although it may result in an elevated decomposition rate under the same environmental conditions due to the increase in the specific surface. Even so, our study can also demonstrate the roles of AOM in regulating Hg methylation in soil and MeHg bioaccumulation in rice. In the future, a controllable cultivation system including a continuous algal growth-death process may infinitely simulate a real environment.

6 Conclusions

This study was conducted to reveal the impacts of algal decomposition in Hg-contaminated paddy fields on MeHg production in soil and its bioaccumulation in rice by adding different dosages of filamentous algae. Algae had a quick decomposition rate, resulting in higher CDOM and water-soluble cysteine concentrations in the soil while lowering soil DOM molecular weight during the rice-tillering period. Compared to other rice growing stages, soil DOM aromatization and humification decreased while water-soluble cysteine concentrations in the soil increased during the grain-filling period. These favorable soil DOM properties and compositions driven by AOM input greatly fueled the MeHg-related regulators, including Hg methylators and non-Hg methylators, and then promoted the MeHg production in the soil. Decayed algal aggregates floating near the soil-water interface acted as the Hg-methylating “hotspot,” as well as the source and sink of soil MeHg. Ultimately, elevated MeHg accumulation in soil regulated by algal decomposition resulted in a great increase in MeHg concentrations and BAF in rice grains. Overall, the input of AOM can promote MeHg synthesis in the soil-water systems by shaping a favorable DOM quality and special microflora, ultimately increasing the MeHg bioaccumulation in rice. Our findings emphasize the environmental implications of algal management in Hg-contaminated paddy fields, worthy of further research.

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Conflict of Interest The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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